

Revisiting Bandlimited Signal Formats for GNSS:

Appendix

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I. Introduction

A. Scope of the document

The scope of this document is to provide a full derivation of the parameters used in [1]. To do so, let us assume that a global navigation satellite system (GNSS) signal $x(t)$ (at the output of the radio frequency (RF) filter, as outlined in [1, Figure 4]) is received embedded in additive white Gaussian noise (AWGN) within a certain observation time window $[0, T_{\text{obs}}]$, whose extension, typically an integer multiple of the spreading sequence period, is much larger than the chip time T_c .

The variance of any estimation algorithm implemented by the GNSS receiver to derive an estimate of the signal time-of-flight (delay) τ is lower bounded by the *modified CRB (MCRB)* [2] (also known as *average CRB*)

$$\text{MCRB}(\tau) = \frac{1}{8\pi^2\beta_x^2 \cdot \text{SNR}}, \quad (1)$$

where $\text{SNR} = P \cdot T_{\text{obs}} / N_0$ is the estimation signal-to-noise ratio (SNR) across the coherent observation window. In the SNR, N_0 is the noise power spectral density (PSD), and the received signal power P , is computed as

$$P = \int_{-B_{\text{rx}}/2}^{B_{\text{rx}}/2} \mathcal{P}_x(f) df, \quad (2)$$

where $\mathcal{P}_x(f)$ denotes the PSD of the transmitted signal $x(t)$. In (1), β_x is the root mean square (rms) bandwidth of the GNSS signal, also called the Gabor bandwidth (GB) of $x(t)$:

$$\beta_x \triangleq \sqrt{\frac{1}{P} \int_{-B_{\text{rx}}/2}^{B_{\text{rx}}/2} f^2 \mathcal{P}_x(f) df}. \quad (3)$$

In (2)-(3), $\mathcal{P}_x(f)$ is derived assuming that the ranging code chips are random. The *modified* appellation in the bound's name indicates that it is calculated under this

same assumption: specifically, it considers the code as a vector of random, independent and identically distributed (iid) *nuisance parameters* [2], [3]. It is seen that both (2) and (3) assume that the receiver processes the GNSS signal within a specific receiver RF bandwidth B_{rx} : as already mentioned, in case the GNSS signal is strictly bandlimited (i.e., it shows a finite RF bandwidth B_x), we assume $B_{\text{rx}} \geq B_x$ (additional considerations along this line will be done later on) and the bound does *not* depend on B_{rx} (it is only dependent on B_x) as the transmitted signal can be fully processed. This corresponds to the practical case when, to achieve higher spectral efficiency and limit out of band emissions, the bandlimiting shall take place at the satellite navigation signal generation unit (NSGU). When we consider instead (theoretically) band-unlimited signals, the received power and the GB do depend on B_{rx} , hence the bound decreases in general as B_{rx} increases, since more and more of the signal spectrum can be processed.

B. Structure of the document

The remainder of the document is structured as follows. After this introduction, Sect. II reports some useful integrals and notation used throughout this technical note. Sect. III investigates the properties of the non-return-to-zero (NRZ) signal, whereas Sects. IV and V analyze the binary offset carrier (BOC) counterparts. Sects. VI and VII tackle the square-root raised cosine (SRRC) signals (without and with introduction of subcarriers). Sects. VIII and IX address the Gaussian and the Gaussian minimum shift keying (GMSK) formats, respectively, whereas Sect. X considers the orthogonal frequency division multiplexing (OFDM) format. Finally, Sect. XI contains some details about realistic channel modeling.

II. Useful results

$$\int_0^x \sin^2(\alpha) d\alpha = \frac{x}{2} - \frac{\sin(2x)}{4} \quad (4)$$

$$\int_0^x \cos^2(\alpha) d\alpha = \frac{x}{2} + \frac{1}{4} \sin(2x) \quad (5)$$

$$\int_0^x \frac{\sin^2(\alpha)}{\alpha^2} d\alpha = \text{Si}(2x) - \frac{\sin^2(x)}{x} \quad (6)$$

$$\int_0^x \frac{\sin^2(\alpha)}{\alpha} d\alpha = \frac{\log \alpha}{2} - \frac{\text{Ci}(2\alpha)}{2} \quad (7)$$

$$\begin{aligned} \int_0^x \alpha \cos^2(\alpha) d\alpha = \\ \frac{x^2}{4} + \frac{1}{4} x \sin(2x) + \frac{1}{8} [\cos(2x) - 1] \end{aligned} \quad (8)$$

$$\begin{aligned} \int_0^x \alpha^2 \cos^2(\alpha) d\alpha = \\ \frac{x^3}{6} + \frac{1}{8} (2x^2 - 1) \sin(2x) + \frac{1}{4} x \cos(2x) \end{aligned} \quad (9)$$

$$\int_0^{+\infty} \alpha^2 \exp\{-\alpha^2\} d\alpha = \frac{\sqrt{\pi}}{4} \quad (10)$$

where

$$\text{Si}(z) \triangleq \int_0^z \frac{\sin(y)}{y} dy, \quad (11)$$

is the integral sine function, and

$$\text{Ci}(z) \triangleq - \int_z^{+\infty} \frac{\cos(y)}{y} dy \quad (12)$$

is the integral cosine function.

III. NRZ

The simplest example of a GNSS signal, assuming no data modulation and unit-power, is expressed by

$$x_N(t) = \sum_{k=-\infty}^{\infty} c[k] p_N(t - kT_N), \quad (13)$$

where $c[k]$ is the k -th chip of the ranging code, T_N is the NRZ chip time, i.e., the inverse of the chiprate R_N , and $p_N(t)$ is a unit-amplitude, T_N -wide rectangular pulse¹ having energy T_N . The corresponding (unit-power) PSD is

$$\mathcal{P}_N(f) = T_N \left[\frac{\sin(\pi f T_N)}{\pi f T_N} \right]^2 = T_N \text{sinc}^2(f T_N). \quad (14)$$

To drive (3), let us use the definition of *normalized* RF receiver bandwidth $W_N = B_{rx} T_N$, and compute separately the numerator

$$\begin{aligned} \int_{-B_{rx}/2}^{B_{rx}/2} f^2 \mathcal{P}_N(f) df &= 2T_N \int_0^{B_{rx}/2} \frac{\sin^2(\pi f T_N)}{\pi^2 T_N^2} df \\ &\stackrel{(a)}{=} \frac{2}{\pi^2 T_N} \int_0^{\pi W_N/2} \sin^2(\nu) \frac{1}{\pi T_N} d\nu \\ &\stackrel{(4)}{=} \frac{2}{\pi^3 T_N^2} \left[\frac{\pi W_N}{4} - \frac{\sin(\pi W_N)}{4} \right] \\ &= \frac{W_N}{2\pi^2 T_N^2} [1 - \text{sinc}(W_N)], \end{aligned} \quad (15)$$

and the denominator

$$\begin{aligned} \int_{-B_{rx}/2}^{B_{rx}/2} \mathcal{P}_N(f) df &= 2T_N \int_0^{B_{rx}/2} \frac{\sin^2(\pi f T_N)}{\pi^2 f^2 T_N^2} df \\ &\stackrel{(a)}{=} 2T_N \int_0^{\pi W_N/2} \frac{\sin^2(\nu)}{\nu^2} \frac{1}{\pi T_N} d\nu \\ &\stackrel{(6)}{=} \frac{2}{\pi} \left[\text{Si}(\pi W_N) - \frac{2 \sin^2(\pi W_N/2)}{\pi W_N} \right] \\ &= W_N \left[\frac{2 \text{Si}(\pi W_N)}{\pi W_N} - \text{sinc}^2(W_N/2) \right], \end{aligned} \quad (16)$$

¹The sans-serif subscript N is used to emphasize that the related quantity is specific of the NRZ pulse – the same notation will be adopted for all of the other signals and related parameters, such as transmitted bandwidth.

where in both cases (a) accounts for the substitution $\nu = \pi f T_N$. Plugging (15) and (16) in (3), we get

$$\beta_N^2 = \frac{1}{2\pi^2 T_N^2} \frac{1 - \text{sinc}(W_N)}{[2 \text{Si}(\pi W_N) / (\pi W_N) - \text{sinc}^2(W_N/2)]}, \quad (17)$$

and finally, using this result in (1), we obtain

$$\frac{\text{MCRB}_N(\tau)}{T_N^2} = \frac{\left[\text{Si}(\pi W_N) + \frac{\cos(\pi W_N) - 1}{\pi W_N} \right]}{2\pi \text{SNR} \left[W_N - \frac{\sin(\pi W_N)}{\pi} \right]}. \quad (18)$$

IV. Sine-phased BOC_s(p, q)

BOC signals with NRZ chips are commonly referred to as BOC(p, q) [4], where the first parameter p defines the sub-carrier rate $f_{sc} \triangleq p f_0$, and the second parameter q defines the ranging code chiprate $R_B \triangleq 1/T_B = q f_0$. The ratio $n = 2f_{sc}/R_B = 2p/q$ represents the number of *half periods* of the square-wave sub-carrier during one code chip time, and can be either odd or even. There exist two versions of BOC signals: a sine-phased version BOC_s(p, q), in which the (odd-symmetric) square-wave subcarrier is obtained by hard-limitation of a sine waveform with the prescribed frequency, and a cosine-phased version, BOC_c(p, q), in which the (even-symmetric) square-wave subcarrier is obtained by hard-limitation of a cosine waveform.

The *sine-phased BOC* signal with sub-carrier frequency $f_{sc} = 1/T_{sc}$ is given by

$$x_{Bs}(t) = \text{sgn}[\sin(2\pi f_{sc} t)] \sum_{k=-\infty}^{\infty} c[k] p_N(t - kT_B), \quad (19)$$

and its PSD turns out to be [5]

$$\mathcal{P}_{Bs}(f, p, q) = \begin{cases} T_B \frac{\sin^2(\pi f T_B)}{(\pi f T_B)^2} \tan^2\left(\frac{\pi f T_B}{n}\right), & n \text{ even} \\ T_B \frac{\cos^2(\pi f T_B)}{(\pi f T_B)^2} \tan^2\left(\frac{\pi f T_B}{n}\right), & n \text{ odd.} \end{cases} \quad (20)$$

To compute (3), let us recall the definition $W_B = B_{rx} T_B$, and let us use (20) to compute separately numerator

$$\begin{aligned} \int_{-B_{rx}/2}^{B_{rx}/2} f^2 \mathcal{P}_{Bs}(f) df &\stackrel{(b)}{=} \\ &= \frac{2n}{\pi^3 T_B^2} \left[\frac{\varphi_0 \pi W_B}{2n} + \sum_{m=1}^n \varphi_m \sin\left(\frac{m\pi W_B}{n}\right) \right], \end{aligned} \quad (21)$$

and denominator

$$\begin{aligned} \int_{-B_{rx}/2}^{B_{rx}/2} \mathcal{P}_{Bs}(f) df &\stackrel{(b)}{=} \\ &= \frac{2n}{\pi} \left[\sum_{m=1}^n \xi_m \text{Si}\left(\frac{m\pi W_B}{n}\right) + \frac{2n}{\pi W_B} \sum_{m=0}^n \psi_m \cos\left(\frac{m\pi W_B}{n}\right) \right], \end{aligned} \quad (22)$$

where in both cases (b) accounts for the substitution $\nu = \pi f T_B/n$, and with the vector coefficients $\varphi = [\varphi_0, \varphi_1, \dots, \varphi_n]$, $\xi = [\xi_1, \dots, \xi_n]$, and $\psi = [\psi_0, \psi_1, \dots, \psi_n]$ that can be obtained by partial integration, and are summarized in Table I for some values of

Table I: Coefficients used to compute (23).

n	1	2	3	4
φ	$[1/2, -1/4]$	$[3/2, -1, 1/8]$	$[5/2, -2, 1/2, -1/12]$	$[7/2, -3, 1, -1/3, 1/16]$
ξ	$[-1, 1]$	$[-1, 1, -1/2]$	$[-8/9, -8/9, 1/3]$	$[-3/4, -1, 3/4, -1/4]$
ψ	$[-1/2, 1/2]$	$[-3/8, 1/2, -1/8]$	$[-5/18, 4/9, -2/9, 1/18]$	$[-7/32, 3/8, -1/4, 1/8, -1/32]$
n	5		6	
φ	$[9/2, -4, 3/2, -2/3, 1/4, -1/20]$		$[11/2, -5, 2, -1, 1/2, -1/5, 1/24]$	
ξ	$[-16/25, -24/25, 24/25, -16/25, 1/5]$		$[-5/9, -8/9, 1, -8/9, 5/9, -1/6]$	
ψ	$[-9/50, 8/25, -6/25, 4/25, -2/25, 1/50]$		$[-11/72, 5/18, -2/9, 1/6, -1/9, 1/18, -1/72]$	

Table II: Coefficients used to compute (29).

n	1	2	3
φ	$[3/2, -2, 1/4]$	$[5/2, -2, -1, 2/3, -1/8]$	$[7/2, -2, -2, 2/3, 1/2, -2/5, 1/12]$
ξ	$[-2, -1]$	$[-1/2, 1, -3/2, 1/2]$	$[-2/9, 8/9, -2/3, -8/9, 10/9, -1/3]$
ψ	$[-3/2, 2, -1/2]$	$[-5/8, 1/2, 1/2, -1/2, 1/8]$	$[-7/18, 2/9, 4/9, -2/9, -2/9, 2/9, -1/18]$
n	4		5
φ	$[9/2, -2, -3, 2/3, 1, -2/5, -1/3, 2/7, -1/16]$		$[11/2, -2, -4, 2/3, 3/2, -2/5, -2/3, 2/7, 1/4, -2/9, 1/20]$
ξ	$[-1/8, 3/4, -3/8, -1, 5/8, 3/4, -7/8, 1/4]$		$[-2/25, 16/25, -6/25, -24/25, 2/5, 24/25, -14/25, -16/25, 18/25, -1/5]$
ψ	$[9/32, 1/8, 3/8, -1/8, -1/4, 1/8, 1/8, -1/8, 1/32]$		$[-11/50, 2/25, 8/25, -2/25, -6/25, 2/25, 4/25, -2/25, -2/25, 2/25, 1/50]$
n	6		
φ	$[13/2, -2, -5, 2/3, 2, -2/5, -1, 2/7, 1/2, -2/9, -1/5, 2/11, -1/24]$		
ξ	$[-1/18, 5/9, -1/6, -8/9, 5/18, 1, -7/18, -8/9, 1/2, 5/9, -11/18, 1/6]$		
ψ	$[-13/72, 1/18, 5/18, -1/18, -2/9, 1/18, 1/6, -1/18, -1/9, 1/18, 1/18, -1/18, 1/72]$		

n for the reader's convenience. Combining (21) and (22), in (3), we get

$$\beta_{Bs}^2 = \frac{1}{\pi^2 T_B^2} \cdot \frac{\frac{\varphi_0 \pi W_B}{2n} + \sum_{m=1}^n \varphi_m \sin\left(\frac{m\pi W_B}{n}\right)}{\sum_{m=1}^n \xi_m \text{Si}\left(\frac{m\pi W_B}{n}\right) + \frac{2n}{\pi W_B} \sum_{m=0}^n \psi_m \cos\left(\frac{m\pi W_B}{n}\right)}, \quad (23)$$

which, using (1), leads to

$$\frac{\text{MCRB}_{Bs}(\tau)}{T_B^2} = \frac{1}{8 \text{SNR}} \cdot \frac{\sum_{m=1}^n \xi_m \text{Si}\left(\frac{m\pi W_B}{n}\right) + \frac{2n}{\pi W_B} \sum_{m=0}^n \psi_m \cos\left(\frac{m\pi W_B}{n}\right)}{\frac{\varphi_0 \pi W_B}{2n} + \sum_{m=1}^n \varphi_m \sin\left(\frac{m\pi W_B}{n}\right)}, \quad (24)$$

where $W_B = B_{rx} T_B$.

V. Cosine-phased BOC_c(p, q)

The *cosine-phased BOC* signal with sub-carrier frequency $f_{sc} = 1/T_{sc}$ is given by

$$x_{Bc}(t) = \text{sgn}[\cos(2\pi f_{sc} t)] \sum_{k=-\infty}^{\infty} c[k] p_N(t - kT_B), \quad (25)$$

and its PSD turns out to be [5]

$$\mathcal{P}_{Bc}(f, p, q) = \begin{cases} T_B \frac{\sin^2(\pi f T_B)}{(\pi f T_B)^2} \frac{\left[\cos\left(\frac{\pi f T_B}{n}\right) - 1\right]^2}{\cos^2(\pi f T_B/n)}, & n \text{ even}, \\ T_B \frac{\cos^2(\pi f T_B)}{(\pi f T_B)^2} \frac{\left[\cos\left(\frac{\pi f T_B}{n}\right) - 1\right]^2}{\cos^2\left(\frac{\pi f T_B}{n}\right)}, & n \text{ odd}. \end{cases} \quad (26)$$

To compute (3), let us recall the definition $W_B = B_{rx} T_B$, and let us use (26) to compute separately numerator

$$\int_{-B_{rx}/2}^{B_{rx}/2} f^2 \mathcal{P}_{Bc}(f) df \stackrel{(b)}{=} \frac{2n}{\pi^3 T_B^2} \left[\frac{\varphi_0 \pi W_B}{2n} + \sum_{m=1}^{2n} \varphi_m \sin\left(\frac{m\pi W_B}{2n}\right) \right], \quad (27)$$

and denominator

$$\int_{-B_{rx}/2}^{B_{rx}/2} \mathcal{P}_{Bc}(f) df \stackrel{(b)}{=} \frac{2n}{\pi} \cdot \left[\sum_{m=1}^{2n} \xi_m \text{Si}\left(\frac{m\pi W_B}{2n}\right) + \frac{2n}{\pi W_B} \sum_{m=0}^{2n} \psi_m \cos\left(\frac{m\pi W_B}{2n}\right) \right], \quad (28)$$

where in both cases (b) accounts for the substitution $\nu = \pi f T_B/n$, and with the vector coefficients $\varphi = [\varphi_0, \varphi_1, \dots, \varphi_{2n}]$, $\xi = [\xi_1, \dots, \xi_{2n}]$, and $\psi = [\psi_0, \psi_1, \dots, \psi_{2n}]$ that can be obtained by partial integration, and are summarized in Table II for some values of n for the reader's convenience. Combining (27) and (28), in (3), we get

$$\beta_{Bc}^2 = \frac{1}{\pi^2 T_B^2} \cdot \frac{\frac{\varphi_0 \pi W_B}{2n} + \sum_{m=1}^{2n} \varphi_m \sin\left(\frac{m\pi W_B}{2n}\right)}{\sum_{m=1}^{2n} \xi_m \text{Si}\left(\frac{m\pi W_B}{2n}\right) + \frac{2n}{\pi W_B} \sum_{m=0}^{2n} \psi_m \cos\left(\frac{m\pi W_B}{2n}\right)}, \quad (29)$$

which, using (1), leads to

$$\frac{\text{MCRB}_{Bc}(\tau)}{T_B^2} = \frac{1}{8 \text{SNR}} \cdot \frac{\sum_{m=1}^{2n} \xi_m \text{Si}\left(\frac{m\pi W_B}{2n}\right) + \frac{2n}{\pi W_B} \sum_{m=0}^{2n} \psi_m \cos\left(\frac{m\pi W_B}{2n}\right)}{\frac{\varphi_0 \pi W_B}{2n} + \sum_{m=1}^{2n} \varphi_m \sin\left(\frac{m\pi W_B}{2n}\right)}. \quad (30)$$

VI. SRRRC

In this case, the resulting unit-power raised-cosine PSD is

$$\begin{aligned} \mathcal{P}_S(f; T_S, \alpha) &= \\ &= \begin{cases} T_S & |f| \leq \frac{1-\alpha}{2T_S} \\ T_S \cos^2 \left[\frac{\pi T_S}{2\alpha} \left(|f| - \frac{1-\alpha}{2T_S} \right) \right] & \frac{1-\alpha}{2T_S} < |f| \leq \frac{1+\alpha}{2T_S} \\ 0 & \text{elsewhere.} \end{cases} \end{aligned} \quad (31)$$

Assuming as usual that $W_S = B_{rx}T_S \geq (1 + \alpha)$ (i.e., getting the whole signal power with no losses), we get

$$\begin{aligned} &\int_{-B_{rx}/2}^{B_{rx}/2} f^2 \mathcal{P}_S(f; T_S, \alpha) df \\ &= 2T_S \int_0^{\frac{1-\alpha}{2T_S}} f^2 df \\ &+ 2T_S \int_{\frac{1-\alpha}{2T_S}}^{\frac{1+\alpha}{2T_S}} f^2 \cos^2 \left[\frac{\pi T_S}{2\alpha} \left(f - \frac{1-\alpha}{2T_S} \right) \right] df \\ &\stackrel{(c)}{=} \frac{(1-\alpha)^3}{12T_S^2} \\ &+ 2T_S \int_0^{\alpha/T_S} \left(\mu + \frac{1-\alpha}{2T_S} \right)^2 \cos^2 \left(\frac{\pi T_S}{2\alpha} \mu \right) d\mu \\ &\stackrel{(d)}{=} \frac{(1-\alpha)^3}{12T_S^2} + 2T_S \int_0^{\pi/2} \left(\frac{2\alpha}{\pi T_S} \right)^2 \nu^2 \cos^2(\nu) \frac{2\alpha}{\pi T_S} d\nu \\ &+ 2T_S \frac{2(1-\alpha)}{2T_S} \int_0^{\pi/2} \frac{2\alpha}{\pi T_S} \nu \cos^2(\nu) \frac{2\alpha}{\pi T_S} d\nu \\ &+ 2T_S \left(\frac{1-\alpha}{2T_S} \right)^2 \int_0^{\pi/2} \cos^2(\nu) \frac{2\alpha}{\pi T_S} d\nu \\ &\stackrel{(9)}{=} \stackrel{(8)}{=} \stackrel{(5)}{=} \frac{(1-\alpha)^3}{12T_S^2} \\ &+ \frac{16\alpha^3}{\pi^3 T_S^2} \left[\frac{\pi^3}{48} + \frac{1}{8} \left(\frac{\pi^2}{2} - 1 \right) \sin \pi + \frac{\pi}{8} \cos \pi \right] \\ &+ \frac{8(1-\alpha)\alpha^2}{\pi^2 T_S^2} \left[\frac{\pi^2}{16} + \frac{\pi}{8} \sin \pi + \frac{1}{8} (\cos \pi - 1) \right] \\ &+ \frac{(1-\alpha)^2 \alpha}{\pi T_S^2} \left[\frac{\pi}{4} + \frac{1}{4} \sin \pi \right] = \frac{1 + \alpha^2 (3 - 24/\pi^2)}{12T_S^2}, \end{aligned} \quad (32)$$

where (c) and (d) account for the substitutions $\mu = f - \frac{1-\alpha}{2T_S}$, and $\nu = \frac{\pi T_S}{2\alpha} \mu$, respectively. Since, by construction, the denominator (signal power) is equal to 1, β_S^2 is equal to (32), which leads to

$$\frac{\text{MCRB}_S(\tau)}{T_S^2} = \frac{3}{2\pi^2 \text{SNR}} \cdot \frac{1}{1 + \alpha^2 (3 - \frac{24}{\pi^2})}. \quad (33)$$

It is easy to see that the SRRRC MCRB is minimized for $\alpha = 0$.

VII. SRRRC signals with subcarriers

In this case, the resulting PSD can be expressed as

$$\begin{aligned} \mathcal{P}_{BL}(f; p, q, \alpha) &= \\ &= \frac{1}{2} [\mathcal{P}_S(f - pf_0; T_{BL}, \alpha) + \mathcal{P}_S(f + pf_0; T_{BL}, \alpha)], \end{aligned} \quad (34)$$

where $\mathcal{P}_S(f; T, \alpha)$ is defined in (31). The total bandlimited offset carrier (BLOC) bandwidth turns out to be

$$B_{RF} = 2f_0 \left[p + q \frac{(1+\alpha)}{2} \right]. \quad (35)$$

To compute (1), let us exploit the spectrum symmetry of (34), and consider the hypothesis $W_{BL} = B_{rx}T_{BL} \geq 2p/q + (1 + \alpha)$, where $T_{BL} = 1/(qf_0)$, to compute separately numerator

$$\begin{aligned} &\int_{-B_{rx}/2}^{B_{rx}/2} f^2 \mathcal{P}_{BL}(f; p, q, \alpha) df \\ &= 2 \int_0^{B_{rx}/2} f^2 \mathcal{P}_{BL}(f; p, q, \alpha) df \\ &= 2 \int_{f_0[p-q\frac{(1+\alpha)}{2}]}^{f_0[p+q\frac{(1+\alpha)}{2}]} f^2 \mathcal{P}_{BL}(f; p, q, \alpha) df \\ &\stackrel{(34)}{=} \int_{-qf_0\frac{(1+\alpha)}{2}}^{qf_0\frac{(1+\alpha)}{2}} (\nu + pf_0)^2 \mathcal{P}_S(\nu; T_{BL}, \alpha) d\nu \\ &\stackrel{(e)}{=} \int_{-qf_0\frac{(1+\alpha)}{2}}^{qf_0\frac{(1+\alpha)}{2}} \nu^2 \mathcal{P}_S(\nu; T_{BL}, \alpha) d\nu \\ &+ p^2 f_0^2 \int_{-qf_0\frac{(1+\alpha)}{2}}^{qf_0\frac{(1+\alpha)}{2}} \mathcal{P}_S(\nu; T_{BL}, \alpha) d\nu \\ &\stackrel{(f)}{=} f_0^2 \left\{ p^2 + q^2 \left[\frac{1 + \alpha^2 (3 - 24/\pi^2)}{12} \right] \right\}, \end{aligned} \quad (36)$$

in which (e) omits the term due to the cross-product, as the integral of an even function in a symmetric interval is 0, and in which (f) exploits (32) and the unit power of $\mathcal{P}_S(\nu; T_{BL}, \alpha)$ over the full signal bandwidth, and denominator

$$\begin{aligned} &\int_{-B_{rx}/2}^{B_{rx}/2} \mathcal{P}_{BL}(f; p, q, \alpha) df \\ &= 2 \int_0^{B_{rx}/2} \mathcal{P}_{BL}(f; p, q, \alpha) df \\ &= 2 \int_{f_0[p-q\frac{(1+\alpha)}{2}]}^{f_0[p+q\frac{(1+\alpha)}{2}]} \mathcal{P}_{BL}(f; p, q, \alpha) df \\ &= 2 \int_{-qf_0\frac{(1+\alpha)}{2}}^{qf_0\frac{(1+\alpha)}{2}} \mathcal{P}_S(\nu; T_{BL}, \alpha) d\nu = 1. \end{aligned} \quad (37)$$

Hence, β_{BL}^2 is equal to (36). Considering that $T_{BL} = 1/(qf_0)$, this leads to

$$\frac{\text{MCRB}_{BL}(\tau)}{T_{BL}^2} = \frac{1}{8\pi^2 \text{SNR} \left\{ \left(\frac{p}{q} \right)^2 + \left[1 + \frac{\alpha^2}{4} \left(1 - \frac{6}{\pi^2} \right) \right] \right\}}. \quad (38)$$

VIII. Gaussian chip pulse

The Gaussian pulse has the property that its Fourier Transform is also a Gaussian function (i.e., it has a Gaussian-shaped frequency spectrum with no sidelobes), and is maximally compact both in the time and frequency domain [6] – for this reason it is often considered in the design of communication formats. A unit-power Gaussian ranging signal has the form

$$x_G(t) = \sum_{k=-\infty}^{\infty} c[k] p_G(t - kT_G; \gamma), \quad (39)$$

where

$$p_G(t; \gamma) = \sqrt{\frac{2}{\pi\gamma^2}} \cdot \exp \left\{ - \left[\frac{t}{\gamma T_G} \right]^2 \right\} \quad (40)$$

and where the chip time/chip rate are as usual $T_G = 1/R_G$. In (39)-(40), γ is a numerical parameter ≥ 0 that regulates the time and spectrum properties of the signal. The PSD of the Gaussian ranging signal turns out to be

$$\mathcal{P}_G(f; \gamma) = \gamma T_G \sqrt{2\pi} \cdot \exp \left\{ -2(\pi\gamma f T_G)^2 \right\}. \quad (41)$$

Let us assume that $B_{rx} \geq B_G$, where B_G , computed as in

$$B_G = \frac{2\sqrt{\ln(10)}}{\pi} \cdot \frac{R_G}{\gamma} \Rightarrow R_G = \frac{\gamma\pi B_G}{2\sqrt{\ln(10)}}, \quad (42)$$

corresponds to the -20 -dB pulse bandwidth, that can easily be derived using (41). This assumption is equivalent to $W_G = B_{rx} T_G \geq 2\sqrt{\ln(10)}/(\pi\gamma)$, considered throughout the paper.

To compute the numerator of (3), let us exploit the symmetry of (41), so that

$$\begin{aligned} & \int_{-B_{rx}/2}^{B_{rx}/2} f^2 \mathcal{P}_G(f; \gamma) df \\ &= 2 \int_0^{B_{rx}/2} f^2 \mathcal{P}_G(f; \gamma) df \stackrel{(g)}{\approx} 2 \int_0^{+\infty} f^2 \mathcal{P}_G(f; \gamma) df \\ &= 2 \int_0^{+\infty} \sqrt{2\pi} \gamma T_G f^2 \exp \left\{ -2(\pi\gamma f T_G)^2 \right\} df \\ &\stackrel{(h)}{=} \frac{1}{\sqrt{\pi}(\pi\gamma T_G)^2} \int_0^{+\infty} \nu^2 \exp \left\{ -\nu^2 \right\} d\nu \\ &\stackrel{(10)}{=} \frac{1}{(2\pi\gamma T_G)^2}, \end{aligned} \quad (43)$$

where (g) accounts for the fact that almost all bandwidth has been considered, based on the hypothesis $B_{rx} \geq B_G$, and (h) performs the substitution $\nu = \sqrt{2}(\pi\gamma f T_G)$. Since, by construction, the denominator (signal power) is equal to 1, β_G^2 is equal to (43). Combining it with (1), we finally get

$$\frac{\text{MCRB}_G(\tau)}{T_G^2} \approx \frac{\gamma^2}{2\text{SNR}} \quad (44)$$

and we can see, as expected, that $\text{MCRB}_G(\tau)$ decreases as γ decreases.

IX. GMSK

Besides the Gaussian pulse illustrated in Sect. VIII, another digital modulation format, often considered when we need to trade off bandwidth occupancy and compactness of the signal amplitude in the time domain, is the minimum shift keying (MSK) format, also including its Gaussian-filtered variants, the GMSK modulation. The baseband (G)MSK signal is given by [6], [7]

$$x_M(t) = \exp \left\{ 2\pi h \sum_{k=-\infty}^{\infty} c[k] q_M(t - kT_M) \right\}, \quad (45)$$

where $c[k]$ is the k -th chip pulse of the ranging code, and, as usual, $T_M = 1/R_M$ is the (G)MSK chip time (the inverse of the chiprate). The coefficient h is the modulation index, usually set to $h = \frac{1}{2}$, and $q_M(t)$ is the phase response

$$q_M(t) = \int_{-\infty}^t g_M(\tau) d\tau, \quad (46)$$

where $g_M(t)$ is the pulse obtained from filtering a T_M -wide rectangular pulse with a Gaussian (low-pass) filter with -3 -dB normalized bandwidth ω_M ,² and conventionally normalized so that its power is $\frac{1}{2}$ (and thus $q_M(+\infty) = \frac{1}{2}$). This format guarantees that the phase of $x_M(t)$ is *continuous* across chip variations (and hence it belongs to the family of continuous-phase modulation (CPM) formats), thus ensuring a theoretical constant signal envelope. Note that, when $\omega_M = +\infty$, the Gaussian filter degenerates into an all-pass filter, and hence $x_M(t)$ (45) is actually a pure MSK format. In what follows, we adopt $\omega_M = 0.3$, to obtain a more compact spectrum.

The PSD $\mathcal{P}_M(f)$ cannot be expressed in a closed form, and hence the MCRB-related quantities. However, there exist many numerical methods (e.g., [7]) to evaluate $\mathcal{P}_M(f)$.

To obtain (1) for the (G)MSK signal, we can first evaluate the autocorrelation function $R_M(\tau)$ as a function of the phase response $q_M(t)$ and considering the chips of the ranging code as equiprobable, using [7, Eq. (17)]. We can then obtain $\mathcal{P}_M(f)$ as the Fourier transform of $R_M(\tau)$, and eventually obtain the GB (3) by numerically integrating $\mathcal{P}_M(f)$, so as to obtain the values of (1).

X. OFDM

A unit-power, symmetric-spectrum OFDM signal is given by

$$x_O(t) = \frac{1}{\sqrt{N_{sc}}} \cdot \sum_{k=0}^{N_{sc}-1} \sum_{\ell=-\infty}^{\infty} c_k[\ell] p_{sc}(t - \ell T_{sc}) e^{j\frac{2\pi\ell}{T_{sc}} \left[k - \frac{N_{sc}-1}{2} \right]}, \quad (47)$$

where N_{sc} is the number of orthogonal active subcarriers, T_{sc} is the *subcarrier chip time*, $1/T_{sc}$ is the spacing

²In the literature, this parameter is usually indicated with β . Here, we use ω_M to avoid conflicts of notation.

between adjacent orthogonal subcarriers, $c_k[\ell]$ is the chip value on subcarrier k (having baseband frequency $(k - N_{\text{sc}}/2 - 1)/T_{\text{sc}}$) at the subcarrier chip time ℓT_{sc} , and $p_{\text{sc}}(t)$ (the subcarrier chip pulse) is an NRZ unit-amplitude pulse with duration T_{sc} . What we call here *subcarrier chiprate* $R_{\text{sc}} = 1/T_{\text{sc}}$ is not commensurate with the chiprate of the other signals presented so far: instead, it represents the rate of the diverse *sub-codes* that we find on any of the subcarriers. Assuming that the OFDM modulator is fed with a “master” ranging code $c[k']$, running at the (master) chiprate $R_{\text{O}} = 1/T_{\text{O}}$, then we have $R_{\text{O}} = N_{\text{sc}}R_{\text{sc}}$ – this is the *master chiprate* that is commensurate to that of the other signals. The relation between the chip index k' of the master code $c[k']$ and the subcarrier/time indices k and ℓ of the subcarriers codes $c_k[\ell]$ is $k' = \ell \cdot N_{\text{sc}} + k$, $0 \leq k \leq N_{\text{sc}} - 1$. In (47), we also neglect for simplicity the presence of a possible cyclic prefix and/or a spectrum shaping filter.

As is known, the OFDM signal is not strictly band-limited. However, when N_{sc} is large, its RF bandwidth is practically equal to $N_{\text{sc}}/T_{\text{sc}} = 1/T_{\text{O}} = R_{\text{O}}$ with a very fast roll-off of the spectrum. Its PSD is in fact given by

$$\begin{aligned} \mathcal{P}_{\text{O}}(f) &= \frac{T_{\text{sc}}}{N_{\text{sc}}} \sum_{k=0}^{N_{\text{sc}}-1} \text{sinc}^2 \left[\left(f - \frac{k}{T_{\text{sc}}} + \frac{N_{\text{sc}}-1}{2T_{\text{sc}}} \right) T_{\text{sc}} \right] \\ &= T_{\text{O}} \sum_{k=0}^{N_{\text{sc}}-1} \text{sinc}^2 [f(N_{\text{sc}}T_{\text{O}}) - (k - (N_{\text{sc}}-1)/2)], \end{aligned} \quad (48)$$

To obtain (1) for the OFDM signal (47), we consider the spectrum (48), and assume that $B_{\text{rx}} \geq \frac{1}{T_{\text{O}}} = \frac{N_{\text{sc}}}{T_{\text{sc}}}$. The denominator in the definition (3) of the (squared) GB is simple to evaluate

$$\begin{aligned} &\int_{-B_{\text{rx}}/2}^{B_{\text{rx}}/2} \mathcal{P}_{\text{O}}(f) df \\ &= \frac{T_{\text{sc}}}{N_{\text{sc}}} \sum_{k=0}^{N_{\text{sc}}-1} \int_{-N_{\text{sc}}/(2T_{\text{sc}})}^{N_{\text{sc}}/(2T_{\text{sc}})} \frac{\text{sinc}^2 \left[\pi f T_{\text{sc}} - \pi \left(k - \frac{N_{\text{sc}}-1}{2} \right) \right]}{\left[\pi f T_{\text{sc}} - \pi \left(k - \frac{N_{\text{sc}}-1}{2} \right) \right]^2} df \\ &\stackrel{(e)}{=} \frac{1}{\pi N_{\text{sc}}} \sum_{k=0}^{N_{\text{sc}}-1} \int_{-\pi(k+1/2)}^{\pi[N_{\text{sc}}-(k+1/2)]} \frac{\text{sinc}^2(\nu)}{\nu^2} d\nu \\ &\stackrel{(6)}{=} \frac{1}{\pi N_{\text{sc}}} \sum_{k=0}^{N_{\text{sc}}-1} \left\{ \text{Si}[\pi(2N_{\text{sc}}-2k-1)] + \text{Si}[\pi(2k+1)] \right. \\ &\quad \left. - \frac{1}{\pi(N_{\text{sc}}-k-1/2)} - \frac{1}{\pi(k+1/2)} \right\}, \end{aligned} \quad (49)$$

where (e) accounts for the substitution $\nu = \pi f T_{\text{sc}} - \pi(k - \frac{N_{\text{sc}}-1}{2})$. We can easily see that, as long as $N_{\text{sc}} \gg 1$, the result of this computation is equal to 1 irrespective of N_{sc} .

Computation of the numerator of (3) is more cumbersome:

$$\begin{aligned} &\int_{-B_{\text{rx}}/2}^{B_{\text{rx}}/2} f^2 \mathcal{P}_{\text{O}}(f) df \\ &= \frac{T_{\text{sc}}}{N_{\text{sc}}} \sum_{k=0}^{N_{\text{sc}}-1} \int_{-N_{\text{sc}}/(2T_{\text{sc}})}^{N_{\text{sc}}/(2T_{\text{sc}})} \frac{\text{sinc}^2 \left[f T_{\text{sc}} - \left(k - \frac{N_{\text{sc}}-1}{2} \right) \right]}{\left[f T_{\text{sc}} - \left(k - \frac{N_{\text{sc}}-1}{2} \right) \right]^2} f^2 df \\ &\stackrel{(e)}{=} \frac{1}{\pi^3 N_{\text{sc}} T_{\text{sc}}^2} \cdot \sum_{k=0}^{N_{\text{sc}}-1} \int_{-\pi(k+1/2)}^{\pi[N_{\text{sc}}-(k+1/2)]} \frac{\text{sinc}^2(\nu)}{\nu^2} \left[\nu + \pi \left(k - \frac{N_{\text{sc}}-1}{2} \right) \right]^2 d\nu \\ &= \frac{1}{\pi^3 N_{\text{sc}} T_{\text{sc}}^2} \sum_{k=0}^{N_{\text{sc}}-1} \int_{-\pi(k+1/2)}^{\pi[N_{\text{sc}}-(k+1/2)]} \text{sinc}^2(\nu) d\nu \\ &\quad + \frac{2}{\pi^2 N_{\text{sc}} T_{\text{sc}}^2} \cdot \sum_{k=0}^{N_{\text{sc}}-1} \left(k - \frac{N_{\text{sc}}-1}{2} \right) \int_{-\pi(k+1/2)}^{\pi[N_{\text{sc}}-(k+1/2)]} \frac{\text{sinc}^2(\nu)}{\nu} d\nu \\ &\quad + \frac{1}{\pi N_{\text{sc}} T_{\text{sc}}^2} \cdot \sum_{k=0}^{N_{\text{sc}}-1} \left(k - \frac{N_{\text{sc}}-1}{2} \right)^2 \int_{-\pi(k+1/2)}^{\pi[N_{\text{sc}}-(k+1/2)]} \frac{\text{sinc}^2(\nu)}{\nu^2} d\nu \\ &\stackrel{(4)}{\stackrel{(7)}{\stackrel{(6)}}{=}} \frac{N_{\text{sc}}}{2\pi^2 T_{\text{sc}}^2} + \frac{1}{\pi^2 N_{\text{sc}} T_{\text{sc}}^2} \cdot \sum_{k=0}^{N_{\text{sc}}-1} \left\{ \left(k - \frac{N_{\text{sc}}-1}{2} \right) \left[\log \left(-1 - \frac{N_{\text{sc}}}{k+1/2} \right) \right. \right. \\ &\quad \left. \left. - \text{Ci}(\pi(2N_{\text{sc}}-2k-1)) + \text{Ci}(\pi(2k+1)) \right] \right\} \\ &\quad + \frac{1}{\pi N_{\text{sc}} T_{\text{sc}}^2} \cdot \sum_{k=0}^{N_{\text{sc}}-1} \left\{ \left(k - \frac{N_{\text{sc}}-1}{2} \right)^2 \left[\text{Si}[\pi(2N_{\text{sc}}-2k-1)] \right. \right. \\ &\quad \left. \left. + \text{Si}[\pi(2k+1)] - \frac{1}{\pi(N_{\text{sc}}-k-\frac{1}{2})} - \frac{1}{\pi(k+\frac{1}{2})} \right] \right\}, \end{aligned} \quad (50)$$

Just like for the case of the denominator, we see that, as long as $N_{\text{sc}} \gg 1$, and according to [8], when the number of equispaced subcarriers is even and the spectrum is symmetric, we can get to the simplification

$$\beta_x^2 \simeq \frac{1}{T_{\text{sc}}^2} \cdot \sum_{k=1}^{N_{\text{sc}}/2} \left(k - \frac{1}{2} \right)^2 = \frac{1}{T_{\text{sc}}^2} \cdot \frac{(N_{\text{sc}}-1)^2}{12}, \quad (51)$$

which eventually leads to

$$\frac{\text{MCRB}_{\text{O}}(\tau)}{T_{\text{sc}}^2} \simeq \frac{3}{2\pi^2} \frac{1}{\text{SNR}(N_{\text{sc}}-1)^2} \simeq \frac{3}{2\pi^2} \frac{1}{\text{SNR} N_{\text{sc}}^2}. \quad (52)$$

Recalling that $T_{\text{sc}} = N_{\text{sc}}T_{\text{O}}$ we also get

$$\frac{\text{MCRB}_{\text{O}}(\tau)}{T_{\text{O}}^2} \simeq \frac{3}{2\pi^2 \text{SNR}} = \frac{\text{MCRB}_{\text{S}}(\tau)}{T_{\text{S}}^2} \Big|_{\alpha=0}. \quad (53)$$

The latter result is pretty natural, since in both cases (OFDM and SRRC with roll-off factor 0) the spectrum of the ranging signal is (approximately or exactly) *rectangular* with the same RF bandwidth.

XI. Realistic system modeling

A. Satellite signal generation

In what follows, we assume a simpler signal than the multichannel one introduced in [1, Sect. III]. In particular, we consider the composition of a unit-power unmodulated pilot $s_p(t)$ on the in-phase (I) component, and a unit-power data-carrying channel $s_d(t)$ on the in-quadrature (Q) component as follows:³

$$\begin{aligned} s(t) &= \sqrt{\frac{2P}{1+\xi}} \left[\sqrt{\xi} s_p(t) + j s_d(t) \right], \\ &= \sqrt{\frac{2P\xi}{1+\xi}} \underbrace{\sum_{k=-\infty}^{\infty} c_p[k] p_{(\cdot)}(t - kT_c)}_{s_p(t)} \\ &\quad + j \cdot \sqrt{\frac{2P}{1+\xi}} \underbrace{\sum_{k=-\infty}^{\infty} c_d[k] d[\lfloor k/L_d \rfloor] p_{(\cdot)}(t - kT_c)}_{s_d(t)}, \end{aligned} \quad (54)$$

where P is the RF signal power, $c_p[k]$ and $c_d[k]$ are the k -th chips of the pseudo noise (PN) ranging sequences, with period M_p and M_d respectively, and used on the pilot and data components, respectively; $d[n]$ is the n -th data symbol, with duration equal to L_d chips; $\lfloor a \rfloor$ denotes the integer part (rounder towards 0) of a ; and $p_{(\cdot)}(t)$ is the time response of the baseband chip shaping pulse, in which the subscript indicates the specific shape following the notation introduced earlier (e.g., $p_N(t)$ is the legacy NRZ pulse). The parameter $0 \leq \xi \leq 1$ finally represents the power ratio between the pilot and data signal components. We will not assume in the present study any forward error-correcting code (FEC) to protect the navigation data $d[n]$.

Our performance analysis in assumes that the GNSS signal is received undistorted at the user terminal and is only made of our pilot component $s_p(t)$ in (54). In this paper, we focus on the contrary on a more realistic configuration of the GNSS signal detection link, as depicted in Figure 1. In our modeling, we assume that the NSGU embeds a linear-phase brick-wall finite impulse response (FIR) digital filter to limit the oversampled digital signal bandwidth to the target signal bandwidth B_x prior digital-to-analog converter (DAC). A possible equiripple implementation of this filter [9] is shown in Figure 2, where the filter RF bandwidth is to $B_x = 20$ MHz, and the out-of-band rejection is -35 dB. In case of SRRC, and in general of strictly bandlimited signals, the brickwall

filter is pointless, since bandlimiting is inherent to signal shaping. We show in Figure 3 shows a possible 257-taps filter implementation to achieve SRRC pulse shaping and very good out-of-band emission.

After the DAC the NSGU embeds an analogue anti-image output filter (not shown in the figure), whose bandwidth is wide enough not to cause any appreciable amplitude and group delay distortion in $[-B_x, B_x]$ while strongly attenuating DAC-generated signal images.

In addition to the digital and analog filters above, we take into account more components that are invariably included in any navigation payload, as shown explicitly in Figure 1: the satellite high power amplifier (HPA) (a linearized travelling wave tube amplifier (LTWTA)), and a satellite (analog) output filter. We will not on the contrary include in our analysis any effects possibly coming from non-ideal antenna behavior. The input signal to the receiver is finally supposed to come from a classical two-ray multipath channel with variable amplitude/phase of the reflected path:

$$y(t) = A\{|x(t)|\} \exp[j\Phi\{|x(t)|\}]. \quad (55)$$

where $A\{\cdot\}$ and $\Phi\{\cdot\}$ denote the amplitude modulation to amplitude modulation (AM-AM) and amplitude modulation to phase modulation (AM-PM) HPA characteristics, respectively. As the satellite HPA we consider an LTWTA, a preferred option for modern GNSS satellites. Thanks to its linearization, the LTWTA exhibits a soft-limiter AM-AM behavior and a reduced AM-PM variation, a feature particularly relevant to limit nonlinear phase distortion on GNSS signals. The measured AM-AM and AM-PM characteristics at L1 used in our simulations are shown in Figure 4.

The payload output filter $h_o(t)$ has the purpose of limiting satellite spurious and harmonics emissions, but at the same time it is designed to introduce very small group delay variations across the useful signal bandwidth. As a consequence, it cannot provide any real signal bandlimitation as it would be desired. This mild filtering, together with the use of NRZ formats, may create significant regulatory issues for the protection of adjacent radio astronomy bands,⁴ as reported in [10]. The amplitude, phase, and group delay responses of the output filter that we used in our simulations (a 4-pole Butterworth filter with one-sided 70-MHz bandwidth) are shown in Figure 5.

B. Propagation channel and receiver noise

The signal $r(t)$ at the input of the receiver is as follows:⁵

$$r(t) = z(t) + \rho \exp(j\theta) z(t - \tau_m) + w(t), \quad (56)$$

³Throughout the paper, all RF signals will be considered through their own I/Q baseband complex-valued equivalent representation.

⁴So much, that in some recent payloads, the band-pass filter is combined with a notch filter to reduce the emissions in a specific adjacent band, at the expense of an increased insertion loss.

⁵We consider possible frequency offsets coming from Doppler effect and demodulator detuning as compensated for.

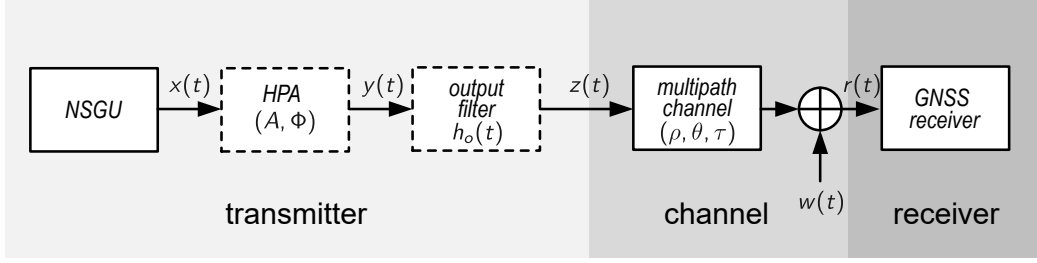


Figure 1: Realistic GNSS system model.

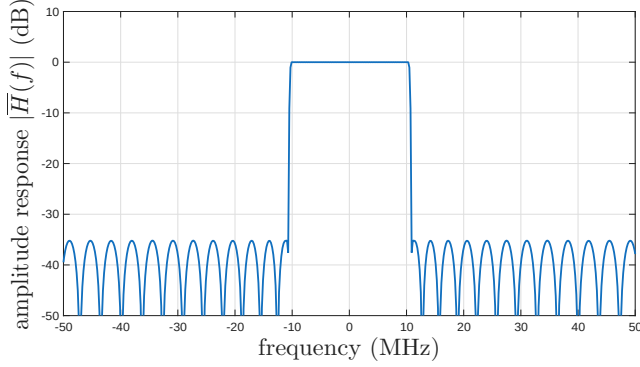


Figure 2: NSGU digital bandlimiting amplitude response for NRZ signal.

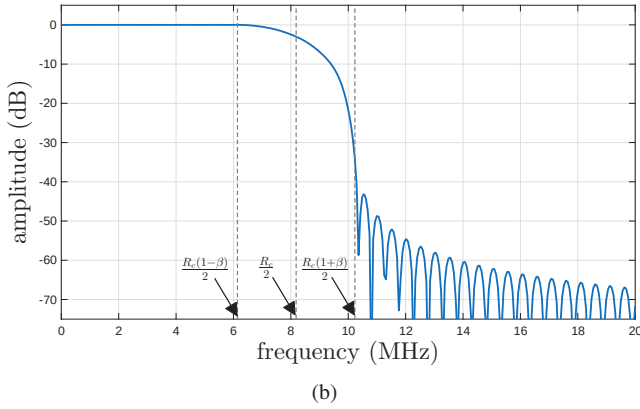
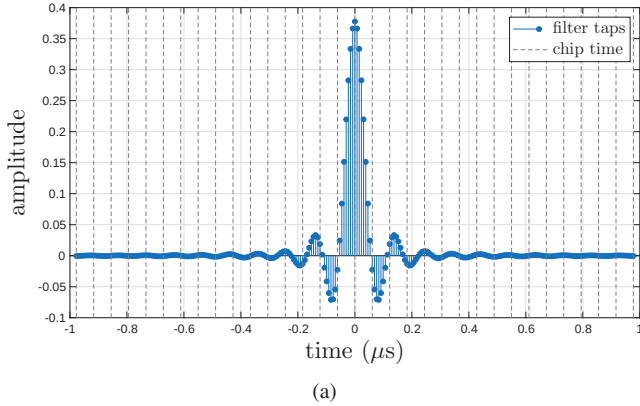


Figure 3: Example of SRRC FIR shaping filter with roll-off $\beta = 0.25$, 257 taps, 8 samples/chips (filter span ± 16 chips): (a) impulse response and (b) frequency response.

where $z(t)$ is the normalized output of the payload in Figure 1; ρ , θ and τ represent the multipath amplitude, phase-shift, and group delay, respectively, and where $w(t)$ is complex-valued AWGN.

C. Receiver signal processing

The general architecture of the GNSS signals demodulator for NRZ and SRRC signals is based on the following main design choices:

- input digital low-pass linear-phase filter with bandwidth B_{rx} (see Figure 2);
- oversampled input signal with respect to (wrt) the Nyquist frequency, commensurate to the input signal relevant bandwidth;
- chip-pulse matched filtering to optimize SNR prior to subsequent processing;
- ideal group delay estimation for replica code alignment prior correlation;
- downsampling to one sample per chip;
- chip-rate correlation with local ranging codes, for both pilot and data signal components;
- accumulation over pilot PN code period and data symbol duration;
- pilot-based phase estimation and correction for data detection;
- individual signal component power estimation after correlation over the ranging code period.

As we will see, the demodulator encompasses some functions (power meter, data demodulation, carrier phase recovery, symbol error rate (SER) estimation) that will be of no relevance to the present study, albeit actually implemented in our simulator.

The main relevant difference between NRZ and SRRC formats is the chip matched filter (CMF), which for NRZ is an integrate-and-dump, and for SRRC is an SRRC filter (same impulse response as the chip pulse shape) to provide an overall raised-cosine inter-chip interference (ICI)-free response with optimum noise bandwidth equal to the chip rate. This demodulator architecture, featuring chip-rate correlations, is typical of low-complexity bandlimited code division multiple access (CDMA) demodulators [11], in that it avoids the need to perform correlation between the incoming signal and the replica code at the oversampled input rate (a feature on the contrary

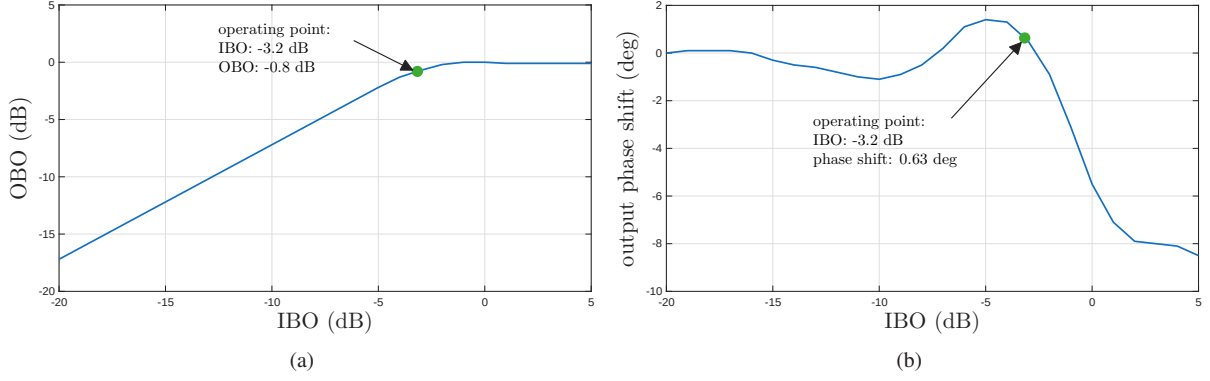


Figure 4: LTWTA satellite payload (a) AM-AM and (b) AM-PM models (measured at L1).

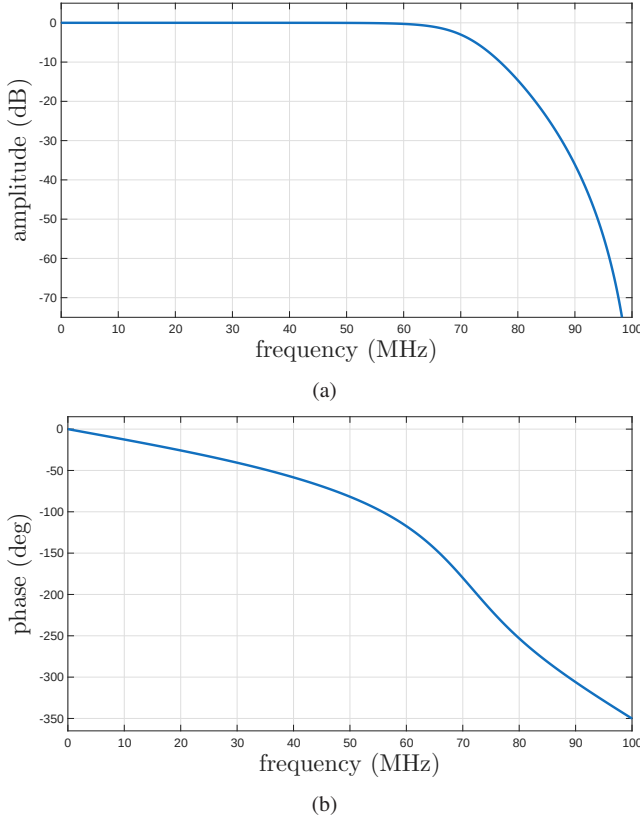


Figure 5: GNSS payload output filter response: (a) amplitude and (b) phase.

very often employed in GNSS demodulators [12]–[14]). Implementing oversampled, NRZ-shaped correlation as is done in classical GNSS receiver may work for NRZ, but leads to some performance degradation for the SRRC signal.⁶

In our simulations, we assumed ideal signal delay estimation: the ideal value can easily be found considering

⁶The use of an SRRC CMF provides extra receiver robustness in the presence of adjacent channel interference. In addition, the filter does not need to be very accurate as, after CMF down-sampling, the samples are accumulated over the PN sequence duration. Hence, residual ICI due to the non perfect filter implementation is averaged out.

the characteristics of the diverse filters and processing blocks. In practice, this function can be performed deriving the early and late samples from the oversampled input signal, together with on-time (prompt) ones, by implementing a modified CMF as described in [15]. Carrier phase estimation based on the pilot channel is on the contrary actually implemented: the nonlinear HPA introduces in general an unknown average phase offset due to the AM/PM characteristics that has to be estimated and compensated for.

D. Implementation aspects

Some aspects related to onboard generation of band-limited (BL) digital signal are addressed in [16], especially the adoption of a software-defined radio (SDR) approach, possibly with direct RF (band-pass) sampling, and no analog frequency conversion [16]. In this architecture, generating a regulated, strictly band-limited signal with multiple channels and/or sub-carriers is only a matter of developing a suitable firmware/software. This approach is also future-proof in the sense that new formats, of course constrained to the same digital bandwidth, can be added by just upgrading the NSGU firmware/software.

Concerning band-limitation, it is true that a strictly band-limited chip pulse cannot be actually generated: in the NSGU, it can be implemented using an FIR shaping filter with a certain number of coefficients (taps). As discussed in Sect. XI.A (see Figure 3), using 129 to 257 coefficients and a sampling frequency of 4 to 8 samples per chip allows us to realize a good approximation of a band-limited pulse with negligible out-of-band frequency components. The SRRC difference to the brick-wall filter currently adopted (see Figure 2) is the different FIR taps value rather than its complexity for similar out-of-band (OOB) emissions.

For the what concerns the on-ground receiving section, focusing for instance on the SRRC format, the receiver first performs SRRC chip matched filtering at the input sampling frequency for all received navigation signals, and, after that, a digital delay lock loop (DDLL) can be implemented with a sampling frequency as low

Parameter	Unit	SiS configuration		
		NRZ	SRRC1	SRRC2
Chip rate/ f_0 ratio		10	16	10
Chip rate	Mcps	10.23	16.368	10.23
Primary code sequence type		Truncated Gold		
Chip pulse shape		Brick-wall filtered NRZ	SRRC	SRRC
SRRC filter roll-off factor α		N/A	0.25	0.25
Allocated SiS bandwidth	MHz	20.46		
Actual SiS bandwidth	MHz	20.460	20.460	12.787
Baseband filter B_{rx}/R_c		2	1.25	1.25
PN spreading sequence period	chips	10, 230	16, 368	10, 230
PN spreading sequence period	ms	1		
Data symbol rate	baud	500		
Power sharing between pilot and data	%	50		

Table III: Study case signal-in-space (SiS) specifications.

as two samples/chip [15] with no performance loss. For each navigation signals, the prompt and early/late samples are taken at chip rate at the output of the chip matched filter using a per-channel interpolator driven by DDLL. The prompt samples are free of ICI (as happens with NRZ pulses) for optimum navigation data detection, and the ranging performance of a noncoherent DDLL lies very close to the relevant MCRB [15]. The demodulator SRRC filter can be realized with a modest number of coefficients, since residual ICI coming from mismatch is mitigated by the subsequent correlation over the (relatively long) PN sequence period. For example, in the laboratory tested implemented by ESA [17] the SRRC demodulator filter selected was optimized to use 10 taps and only requiring 8 additions and 1 subtraction for a loss of 0.17 dB. The effect of signal quantization are not considered here, but have been in-depth investigated for direct-sequence spread-spectrum (DS-SS) bandlimited CDMA signals in [18]. It was found that 4 bits quantization allow us to preserve the spreading sequence orthogonality, but in practice lower number of bits can be used.

E. Satellite payload effects: A study case

To analyze the satellite payload impact, we focus on a specific case study. In particular, we assume an overall bandwidth allocation of 20.4 MHz, and we comply with this requirement including in the NSGU the digital brick wall filter whose response is shown in Figure 2. As for the signal format, we concentrate on *i)* an NRZ 10.23 Mcps/s fitting the given bandwidth at the first null, *ii)* a bandlimited SRRC $\alpha = 0.25$ signal with chip rate 16.368 Mcps/s, and *iii)* a second $\alpha = 0.25$ SRRC format with the same 10.23 Mcps/s NRZ chip rate to further reduce the OOB emissions. The GNSS signal is formatted according to (54), and Table III reports a detailed description of the parameters for the different configurations.

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